01 / PROBLEM AND CONTRIBUTION

Telescopic-cascode amplifier

EEE 523 · Advanced Analog IC Design | Spring 2026 | Individual design

RESULT AT A GLANCE

Characterized a telescopic-cascode amplifier at 3 V supply and 1.5 V common mode, reporting 90.21 dB CMRR, 34.07 dB PSRR+, and 46.75 dB PSRR- at the documented readings.

The engineering question

How do common-mode voltage and transistor bias conditions affect the behavior of a telescopic-cascode differential amplifier?

Situation

A telescopic stack depends on available voltage headroom and the common-mode operating point. The project examines the differential amplifier together with its current reference and bias circuitry.

Task

Establish the transistor operating points and characterize gain, common-mode behavior, supply rejection, and transient response.

My contribution

Differential-stage design, transistor-level biasing, and performance characterization in EEE 523.

Tools and concepts

Analog IC design / Cascode amplifiers / Biasing / PSRR / CMRR / Cadence Virtuoso

02 / DESIGN AND METHOD

From the task to implementation

01 / Implementation step

Combined a beta-multiplier reference, wide-swing cascode biasing, and a telescopic differential stage. Documented transistor dimensions and operating-point parameters.

02 / Implementation step

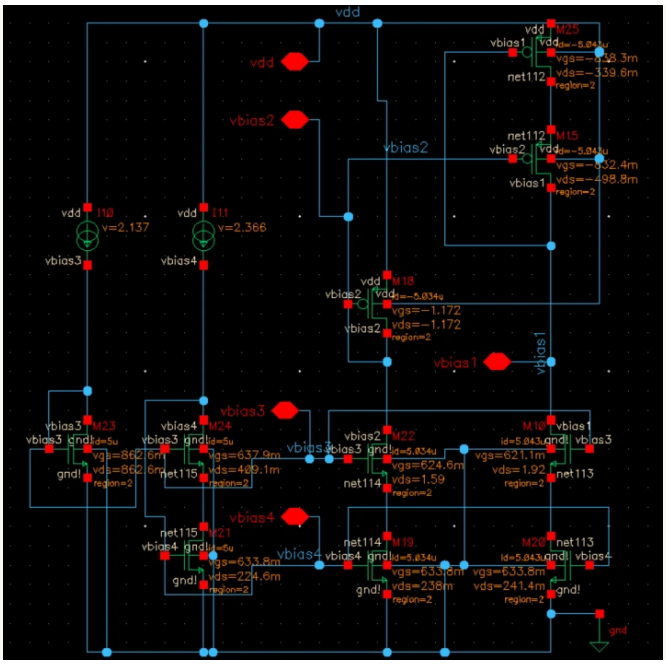
Swept input common-mode voltage and examined gain and phase. Separate testbenches characterized positive- and negative-supply rejection and common-mode rejection.

03 / Implementation step

Configured the amplifier as a unity-gain follower to examine transient response and positive and negative slew behavior.

Biasing and headroom

The beta-multiplier reference and wide-swing cascode bias circuitry support the differential stack. Sweeping common-mode voltage checks how that operating point affects the available headroom and small-signal behavior.



Telescopic amplifier schematic

Transistor-level circuit from the telescopic-cascode report.

03 / RESULTS AND ENGINEERING PERSPECTIVE

Evidence, interpretation, and next steps

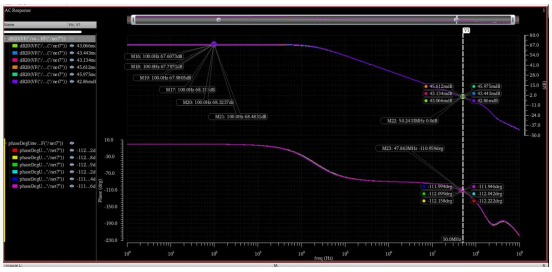
Supply voltage	3 V
Nominal VCM	1.5 V
Reported CMRR	90.21 dB

Reported result

At the report's nominal VDD = 3 V and VCM = 1.5 V condition, the listed rejection values are PSRR+ = 34.07 dB, PSRR- = 46.75 dB, and CMRR = 90.21 dB.

Separate rejection mechanisms

CMRR evaluates the common-mode input path; PSRR+ and PSRR- evaluate disturbance coupling from the two supply directions. The listed dB values belong to the reported testbench readings, not a guaranteed rejection floor over all frequencies.



AC response

Gain and phase plot from the project report.

Scope and limitations

Supply and common-mode rejection depend on frequency and bias conditions. The values above are reported readings at the nominal setup; the broader study also explores common-mode and transient behavior.

Possible next steps - proposed, not completed results

• Tabulate rejection versus frequency and common-mode voltage alongside the nominal readings.

• Extend the bias and transient checks across clearly specified load and process conditions.

Source material: EEE 523 Project2_AAIC.pdf, pp. 3-16.

Prepared from the supplied reports, project files, and verified portfolio copy. Figures retain their original content; explanatory diagrams are labelled.